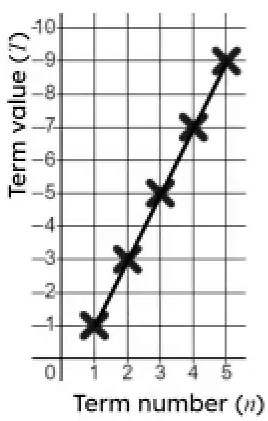
# Checking and securing understanding of plotting coordinates generated from a rule

**1** This table of values tells us the coordinates to plot the relationship $y = 3x + 4$. Fill in the blank

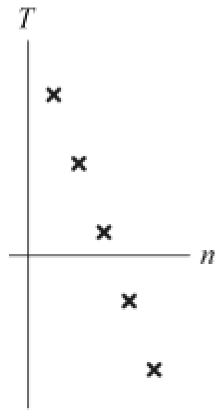
| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | $-5$ | $-2$ | $1$ | $4$ | $7$ | $10$ | $13$ |

**2** What is wrong with this plotting of the arithmetic sequence $2n - 1$? Tick **1** correct answer




- [ ] The coordinates are wrong.
- [ ] The line should extend to the edges of the grid.
- [x] There shouldn't be a line through the points.
- [ ] The points should not line up.
- [ ] You need more than $5$ term numbers.

**3** Which of these arithmetic sequences could this graph represent? Tick **1** correct answer




- [x] $17 - 5n$
- [ ] $5n - 17$
- [ ] $n - 17$
- [ ] $5 - n$
- [ ] $5 - 17n$

**4** What are the two missing $y$ coordinate values for this linear rule? Tick **2** correct answers

| | | | | | | |
|---|---|---|---|---|---|---|
| $x$ | $-2$ | $-1$ | 0 | 1 | 2 | 3 |
| $y$ | | | 4 | 9 | 14 | 19 |

- [x] $-1$
- [ ] $-4$
- [ ] $-5$
- [x] $-6$
- [ ] $-9$

**5** This is a table of values for a linear rule but there is an error. Which value is the error? Tick **1** correct answer

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $x$ | $-3$ | $-2$ | $-1$ | 0 | 1 | 2 | 3 |
| $y$ | 26 | 19 | 12 | 5 | $-3$ | $-9$ | $-16$ |

- [ ] $y = 26$
- [ ] $y = 19$
- [ ] $y = 5$
- [x] $y = -3$
- [ ] $y = -16$

6 For the linear rule $y = 13 - 2x$ what is the $y$ coordinate when $x = -5$? Tick **1** correct answer

- [x] 23
- [ ] 18
- [ ] 11
- [ ] 8
- [ ] 3